



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-22420

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Mon, Jul 29, 2024 at 2:16 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

29-Jul-2024

Dear Mr. Khare:

The manuscript entitled "Finding the Optimal Security Policies for Autonomous Cyber Operations with Competitive Reinforcement Learning" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access

ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Finding the Optimal Security Policies for Autonomous Cyber Operations with Competitive Reinforcement Learning

AUTHORS: MCDONALD, GARRETT; LI, LI; MALLAH, RANWAAL

ABSTRACT: Reinforcement Learning (RL) has been responsible for some of the most impressive advances in the field of Artificial Intelligence (AI). Research in competitive RL has shown that multiple agents competing in an adversarial environment can learn simultaneously in order to discover their optimal decision-making policies. Competitive RL algorithms have been used to train performant AI for a variety of games and optimization problems. Cybersecurity is a domain where the emerging research in competitive RL is being considered for its real-world application. In order to develop Automated Cyber Operations (ACO) tools using RL, various open-source environments are available to simulate network security incidents. However, the existing research in these environments is typically one-sided: a Red or Blue agent is trained to optimize their decision-making against a static opponent. Competitive RL has not been attempted in these emerging environments. In this work, we trained agents using competitive RL to approximate their game theory optimal policies in a simulated ACO environment. We showed that near-optimal behavior was reached gradually through fictitious play demonstrating that these strategies can be used to approximate the optimal policies for agents involved in sophisticated sequential decision-making during a cyber attack.

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